higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## PROCESS TECHNOLOGY
## NQF LEVEL 2

## 18 NOVEMBER 2019

**This marking guideline consists of 6 pages.**

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## SECTION A

### QUESTION 1

1.1 C
1.2 B
1.3 A
1.4 B
1.5 D

(5 × 1) **[5]**

### QUESTION 2

2.1 True
2.2 False
2.3 False
2.4 True
2.5 True

(5 × 1) **[5]**

### QUESTION 3

3.1 natural
3.2 bacteria or micro-organisms
3.3 vibrations
3.4 round
3.5 decks

(5 × 1) **[5]**

**TOTAL SECTION A: 15**

## SECTION B

## QUESTION 4

4.1
- Help operators to understand how a process work
- Trace a process through the plant
- Identify a process or stream
- Keep track of piping and equipment in a processing plant (Any 3 × 1) (3)

4.2 4.2.1 Equipment number (1)

4.2.2 Line number (1)

4.3 pH scale indicates the acidity and alkalinity of a liquid.✓✓ It works on a measure of 0 to 14.✓ If the pH is below 7, it indicates acidity,✓ and if it is above 7, it indicates alkalinity.✓ (5)

4.4 By covering it with a metal or cloth-like cover called lagging (2)

4.5
- Hand wheel
- Handle
- Lever (3)

**[15]**

## QUESTION 5

5.1 A – Discharge line
B – Casing
C – Volute
D – Suction eye
E – Impeller
F – Suction line (6)

5.2 5.2.1
- Piston pump
- Diaphragm pump (2)

5.2.2 They transfer fluids by means of direct pressure and not centrifugal force.✓ They have valves for the intake and release of liquids.✓ A volume of liquid is drawn into the cylinder by the suction on the intake stroke✓ and is discharged through the outlet valves on the discharge stroke.✓ The discharge from the pump is pulsating and can be changed only by changing the speed of the pump.✓ This is because the intake is always at constant volume.✓ (6)

5.3 It is a gas that contains moisture droplets. (2)

5.4 The downward motion of the piston causes sufficient initial pressure to open the cup and produce a tight seal between the leather cup and the seal wall.✓Air is forced through a non-return valve into the tyre.✓ The upward motion of the piston creates a partial vacuum inside the cylinder,✓ permitting atmospheric air to flow past the leather cup and fill the cylinder so that the cycle may be repeated.✓ (4)

**[20]**

## QUESTION 6

6.1
- To ensure the availability of raw materials and intermediate products
- To store end products before distribution (2)

6.2 Storage of large quantities of solids, liquids or gases (2)

6.3
- Vessels
- Tanks
- Containers
- Silos
- Bulk bags
- Bunkers
- Hoppers
- Bins (Any 5 × 1) (5)

6.4
- Raw materials
- Final products
- By-products
- Intermediate products (4)

6.5 Containers that store liquid in bulk (2)

6.6 6.6.1 To prevent the contamination of the environment (2)

6.6.2 In order to be protected from dangerous elements (2)

6.6.3 They will rust. (1)

**[20]**

## QUESTION 7

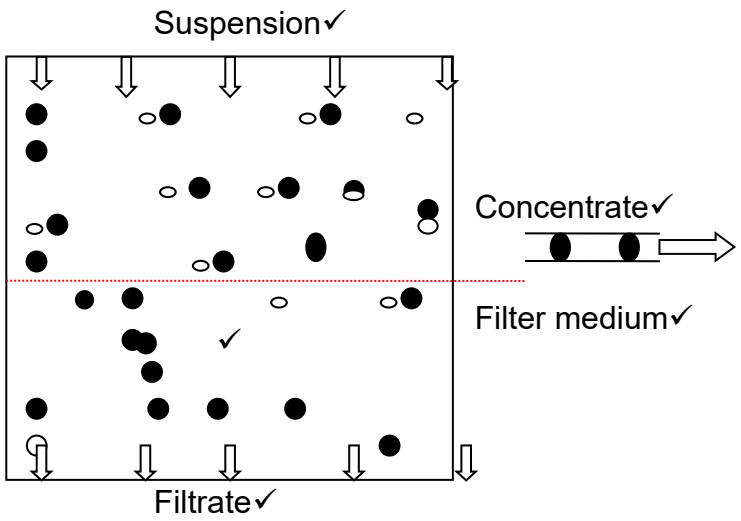
7.1 7.1.1 The fluid that has passed through the filter or the fluid that has been filtered

7.1.2 The concentrated mixture of solids rejected by the filters

7.1.3 The haziness of water caused by suspended particles

(3 × 2) (6)

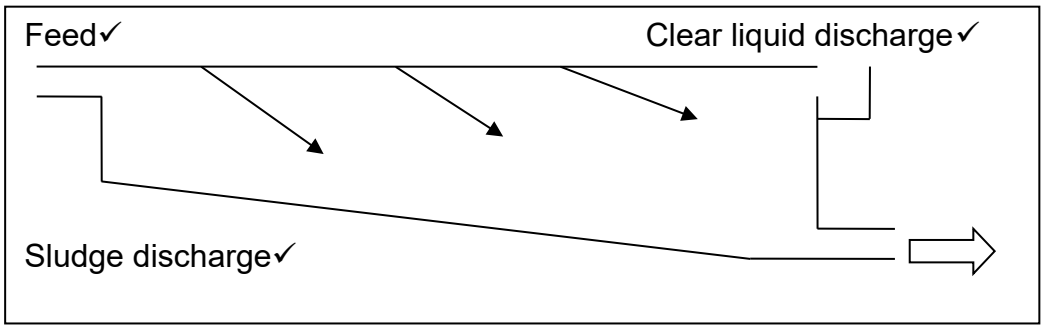
7.2




(5)

7.3 The suspension is fed to the tank, from where the liquid passes through the sand✓and the particulate solids are trapped within the sand.✓ Later a dirt blanket is formed on the top of the sand.✓ The dirt blanket removes turbidity of the water in the water treatment process.✓ After the filter has operated for a few weeks, the top layer becomes clogged✓ and filtration becomes very slow.✓ The top layer of the sand is removed from the filter to keep it clean.✓ The process requires manual cleaning, as the filter lacks the backwashing equipment used in the rapid sand filter.✓ (8)

7.4 Working principles of a rectangular thickener or settling tank




The liquid flows in on the one side and is removed on the other side.✓ The tank gets deeper towards the far end,✓ where the sludge is collected and removed from the tank.✓ (6)

7.5 To remove low-density materials, air and other gases from the feed stream (2)

7.6
- Cooling crystallisers
- Evaporation crystallisers
- Adiabatic cooling crystallisers (3)

**[30]**

**TOTAL SECTION B: 75**
**GRAND TOTAL: 100**

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